MTH108 JUL21 TOA Review

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Question 1

(a) Let $f(x) = e^{x-2} - 2$, defined over the domain $x \leq 1$. Determine an expression for $f^{-1}(x)$, as well as its domain and range.

(7 marks)

(b) Determine the derivative of $g(x) = \sqrt{1-x^2}\,\sin^{-1}(x)$ for $-1 < x < 1$, simplifying your answer as much as possible.

(6 marks)

(c) For the function $h(x) = \log_{2021}\left(\frac{x^{2021}}{2021^x}\right)$, determine the exact value of $h'(2021)$.

(7 marks)

(a)

domain of $f(x) = (-\infty, 1]$

$x \leq 1 \qquad e^{x-2} \leq e^{-1}$

$x-2 \leq -1 \qquad e^{x-2}-2 \leq e^{-1}-2$

$\therefore f(x) \leq e^{-1}-2$

range of $f(x) = (-\infty, e^{-1}-2]$

let $y = e^{x-2}-2$

$y+2 = e^{x-2}$

$\ln(y+2) = x-2$

$x = 2+\ln(y+2)$

$\therefore f^{-1}(x) = 2+\ln(x+2)$

domain of f^{-1} = range of f

$= (-\infty, e^{-1}-2]$

range of f = domain of f

$= (-\infty, 1]$

(b) $g'(x) = \frac{d}{dx}\sqrt{1-x^2}\,\sin^{-1}x$

$= \left[\frac{1}{2}(1-x^2)^{-\frac{1}{2}}(-2x)\right]\sin^{-1}x + \sqrt{1-x^2}\left[\frac{1}{\sqrt{1-x^2}}\right]$

$= 1 - \frac{x}{\sqrt{1-x^2}}$

(c) $h(x) = \log_{2021}\left[\frac{x^{2021}}{2021^x}\right]$ $\qquad \log\frac{a}{b} = \log a - \log b$

$= \log_{2021} x^{2021} - \log_{2021} 2021^x$

$= 2021\log_{2021} x - x$

$\log_b a = \frac{\log_c a}{\log_c b}$

$= 2021\left[\frac{\ln x}{\ln 2021}\right] - x$

$h'(x) = \frac{2021}{\ln 2021}\cdot\frac{1}{x} - 1$

$h'(2021) = \frac{2021}{\ln 2021}\cdot\frac{1}{2021} - 1$

$= \frac{1}{\ln 2021} - 1$

Question 2

Use various techniques of integration to evaluate the following integrals.

(a) $\int \frac{\cos x}{(\sin x)^2 + 3\sin x}\, dx$

(8 marks)

(b) Determine the exact value of $\int_1^2 3xe^x\, dx$.

(8 marks)

(a) $\int \frac{\cos x}{\sin^2 x + 3\sin x}\, dx$ let $du = \cos x\, dx$ $\Rightarrow u = \sin x$

$= \int \frac{1}{u^2+3u}\, du$

$= \int \frac{1}{u(u+3)}\, du$ convert to partial fraction

$= \frac{1}{3}\int \frac{1}{u} - \frac{1}{u+3}\, du$

$= \frac{1}{3}\left[\ln|u| - \ln|u+3|\right] + k$

$= \frac{1}{3}\ln|\sin x| - \frac{1}{3}\ln|\sin x+3| + k$

where k is a constant

(b) $\int 3xe^x dx = \int \underbrace{(3x)}_{u}\, \underbrace{e^x}_{dv}\, dx$

$= 3xe^x - \int 3e^x\, dx$

$= 3xe^x - 3\int e^x\, dx$

$= 3xe^x - 3e^x + k$

where k is a constant

$\int_1^2 3xe^{3x} dx = \left[3xe^x - 3e^x\right]_1^2$

$= 3\left[(x-1)e^x\right]_1^2$

$= 3\left[e^2 - 0\right]$

$= 3e^2$

Question 3

(a) Interpret $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{\frac{5}{4n}}{1+\frac{5k}{4n}}$ as a definite integral and evaluate it.

(8 marks)

(b) Let $f(x) = \int_1^{x^2} \frac{1+\sin t}{\sqrt{t}}\, dt$. Solve for $f'(x)$ and show that $f'(x) \le 4$ for every $x > 0$.

(8 marks)

(a) $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{\frac{5}{4n}}{1+\frac{5k}{4n}}$

$b-a=\frac{1}{4}$, $b=\frac{1}{4}$

$x_k = a + k\Delta x$

$= \lim_{n\to\infty} \sum_{k=1}^{n} \frac{\frac{1}{4}}{n} \cdot \frac{5}{1+5[k(\frac{1}{4n})+0]}$

$a=0$

x_k

$= \lim_{n\to\infty} \sum_{k=1}^{n} \frac{\frac{1}{4}}{n} \cdot f(k(\frac{1}{4n})+0)$ where $f(x) = \frac{5}{1+5x}$

$= \int_0^{\frac{1}{4}} f(x)\,dx = \int_0^{\frac{1}{4}} \frac{5}{1+5x}\,dx$

$= \int_0^{\frac{1}{4}} \frac{5}{1+5x}\,dx$

$= \left[\ln|1+5x|\right]_0^{\frac{1}{4}}$

$= \ln\frac{9}{4}$

(b) let $g(t) = \frac{1+\sin t}{\sqrt{t}}$

By FTC2

$\int_1^{x^2} g(t)\,dt = G(x^2) - G(1)$

$\frac{d}{dx}\int_1^{x^2} g(t)\,dt = \frac{d}{dx}\left[G(x^2) - G(1)\right]$

$= g(x^2)\frac{d}{dx}(x^2) - 0$

$= \frac{1+\sin x^2}{x} \cdot 2x$

$= 2 + 2\sin x^2$

$\therefore f'(x) = 2 + 2\sin x^2$

$-1 \le \sin x^2 \le 1$

$-2 \le 2\sin x^2 \le 2$

$2-2 \le 2+2\sin x^2 \le 2+2$

$0 \le 2+2\sin x^2 \le 4$

$\therefore f'(x) \le 4$

(c) Show that the following reduction formula holds, by using integration by parts:

$$\int \frac{e^{mx}}{x^n}\,dx = -\frac{e^{mx}}{(n-1)\,x^{n-1}} + \frac{m}{n-1}\int \frac{e^{mx}}{x^{n-1}}\,dx\,,$$

where $m, n \in \mathbb{Z}$ and $n > 1$.

(8 marks)

3(c) $\int \frac{e^{mx}}{x^n}\,dx = \int \underbrace{e^{mx}}_{u}\,\underbrace{x^{-n}}_{dv}\,dx$ we can obtain x^{-n} by integrating x^n

$$= \frac{e^{mx}x^{1-n}}{1-n} - \int (me^{mx})\frac{x^{1-n}}{1-n}\,dx$$

$$= -\frac{e^{mx}}{(n-1)x^{n-1}} + \int \frac{me^{mx}}{(n-1)x^{n-1}}\,dx$$

$$= -\frac{e^{mx}}{(n-1)x^{n-1}} + \frac{m}{n-1}\int \frac{e^{mx}}{x^{n-1}}\,dx$$

Question 4

Let R be the region in the first quadrant bounded by the curves $y = x^2 + 1$ and $y = 1 + 4x - x^2$. [A sketch of the region might be helpful.]

(a) Find the area of the region R.

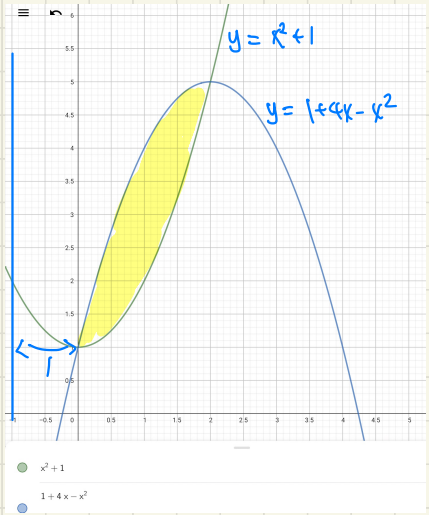
(6 marks)

(b) Using the disk/washer method, find the volume of the solid of revolution obtained when R is rotated 2π radians about the x-axis.

(7 marks)

(c) Using the method of cylindrical shells, find the volume of the solid of revolution obtained when R is rotated 2π radians about the line $x = -1$.

(7 marks)



(a)

let $1+4x-x^2 = x^2+1$

$2x^2-4x=0$

$2x(x-2)=0$

$x=0$ or 2

$y=1$ or 5

find intersection

$$\text{area} = \int_0^2 ((1+4x-x^2)-(x^2+1))\,dx$$

$$= \int_0^2 4x-2x^2\,dx$$

$$= \left[2x^2-\tfrac{2}{3}x^3\right]_0^2$$

$$= 8 - \frac{16}{3}$$

$$= \frac{8}{3}$$

(b) $r_1(x) = 1+4x-x^2$

$r_2(x) = x^2+1$

$$\text{volume} = \int_0^2 \pi[r_1(x)]^2 - \pi[r_2(x)]^2\,dx$$

$$= \pi\int_0^2 (1+4x-x^2)^2 - (x^2+1)^2\,dx$$

$$= \pi\int_0^2 -8x^3+12x^2+8x\,dx$$

$$= \pi\left[-2x^4+4x^3+4x^2\right]_0^2$$

$$= 16\pi$$

(c) $r(x) = x-(-1) = x+1$

$h(x) = (1+4x-x^2)-(x^2+1) = 4x-2x^2$

$$\text{volume} = \int_0^2 2\pi r(x)h(x)\,dx$$

$$= 2\pi\int_0^2 (x+1)(4x-2x^2)\,dx$$

$$= 2\pi\int_0^2 -2x^3+2x^2+4x\,dx$$

$$= 2\pi\left[-\tfrac{1}{2}x^4+\tfrac{2}{3}x^3+2x^2\right]_0^2$$

$$= 2\pi\left(\frac{16}{3}\right)$$

$$= \frac{32\pi}{3}$$

Question 5

(a) Evaluate the following integral, showing all steps of your working clearly:

$$\int \tan^{-1} x \, dx.$$

(8 marks)

(b) Hence, or otherwise, determine the exact value of

$$\int_0^1 (\tan^{-1} x)\left(\frac{x^2}{1+x^2}\right) dx.$$

(12 marks)

5(a) $\int \tan^{-1} x \, dx = \int \overset{dv}{(1)} \overset{u}{\tan^{-1} x} \, dx$

$= x \tan^{-1} x - \int x \frac{1}{x^2+1} dx$

$= x \tan^{-1} x - \frac{1}{2} \int \frac{2x}{x^2+1} dx$

$= x \tan^{-1} x - \frac{1}{2} \ln(x^2+1) + C,$

where C is a constant

(b) $\int (\tan^{-1} x)\left(\frac{x^2}{1+x^2}\right) dx$

$= \int (\tan^{-1} x)\left(\frac{x^2+1-1}{1+x^2}\right) dx$

$= \int (\tan^{-1} x)\left(1 - \frac{1}{1+x^2}\right) dx$

$= \int \tan^{-1} x - \frac{\tan^{-1} x}{1+x^2} dx$

$f'(x) \cdot f(x) \Rightarrow 2 f'(x) f(x) \Rightarrow \frac{d}{dx}[f(x)]^2$

'reverse' chain rule

$= \int \tan^{-1} x \, dx - \frac{1}{2} \int 2\left(\frac{1}{1+x^2}\right) \tan^{-1} x \, dx$

$= \int \tan^{-1} x \, dx - \frac{1}{2} (\tan^{-1} x)^2 + C$

$\int_0^1 (\tan^{-1} x) \frac{x^2}{1+x^2} dx$

$= \int_0^1 \tan^{-1} x \, dx - \frac{1}{2} \left[(\tan^{-1} x)^2\right]_0^1$

$= \left[x \tan^{-1} x - \frac{1}{2} \ln|x^2+1| - \frac{1}{2} (\tan^{-1} x)^2\right]_0^1$

$= \frac{\pi}{4} - \frac{1}{2} \ln 2 - \frac{1}{2}\left(\frac{\pi}{4}\right)^2$